National University of Computer and Emerging Sciences
Lahore Campus

Linear Algebra (MT1004) Final Exam

Date: May 27th 2024

Course Instructor(s)

Dr. Hira Iqbal

Total Time (Hrs): 3

Total Marks: 70

Total Questions: 5

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Roll No Section Student Signature

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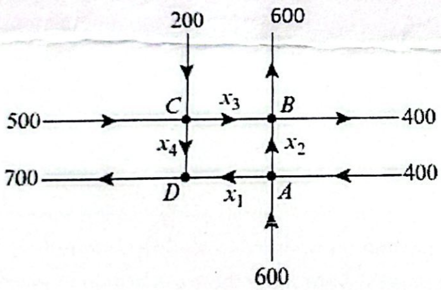
Attempt all the questions. Ensure that parts of the same question are attempted together.

CLO 1: Express real-life problems into system of linear equations and solve the system.

Q1: The accompanying figure shows a network of one-way streets with traffic flowing in the directions indicated. The flow rates along the streets are measured as the average number of vehicles per hour [10 marks]



Set up a linear system whose solution provides the unknown flow rates and solve the system for the unknown flow rates.

CLO 2: Recognize vector spaces and/or subspaces and develop their bases.

Q2: [5+5 marks]

a) Determine whether the set of all vectors of the form (a, b, c), where $b = a + c + 1$ forms a subspace in R^3.

b) Compute the basis for the column space of the matrix

$$A = \begin{bmatrix} 1 & -2 & 5 & 0 & 3 \\ -2 & 5 & -7 & 0 & -6 \\ -1 & 3 & -2 & 1 & -3 \\ -3 & 8 & -9 & 1 & -9 \end{bmatrix}$$

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CLO 2: Recognize vector spaces and/or subspaces and develop their bases.

Q3: Calculate the eigenbases of A and then determine the matrix P that diagonalizes A. [10 marks]

$$A = \begin{bmatrix} 2 & 0 & -2 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{bmatrix}$$

CLO 3: Identify inner product spaces and construct orthogonal/orthonormal basis.

Q4: [5+5+10 marks]

a) Let $u = (u_1, u_2, u_3)$ and $v = (v_1, v_2, v_3)$. The expression $< u, v > = u_1v_1 + u_2v_2 - u_3v_3$ does not define an inner product on R^3. Which axioms of inner product $< u, v >$ fails to hold?

b) The vectors $v_1 = \left(-\frac{3}{5}, \frac{4}{5}, 0\right), v_2 = \left(\frac{4}{5}, \frac{3}{5}, 0\right), v_3 = (0,0,1)$ form an orthonormal basis for R^3 with respect to the Euclidean inner product, express the vector $u = (3, -7, 4)$ as a linear combination of v_1, v_2, and v_3.

c) *Let R^3 have the Euclidean inner product and apply the Gram–Schmidt process to transform the basis $\{u_1, u_2, u_3\}$ into an orthogonal basis where $u_1 = (1, 1, 1), u_2 = (-1, 1, 0), u_3 = (1, 2, 1)$*

CLO 4: Express a linear transformation graphically/ as matrices and use their properties to solve engineering problems.

[10+10 marks]

Q5:

a) An image with coordinates (1, 3), (0, 4), (−1 ,3) is obtained by transforming the coordinates of the original image first by shearing it in the x direction by a factor 2, call it T_1 and then by reflecting it about the line $y = x$, call it T_2. Determine the transformation matrix $T_2 o\ T_1$ and then compute the inverse transformation that retrieves the original image. What are the original coordinates?

b) Consider the basis $S = \{v_1, v_2\}$ for R^2, where $v_1 = (0, 1)$ and $v_2 = (1, 1)$, and let $T : R^2 \rightarrow R^2$ be the linear operator for which $T(v_1) = (-1, -2)$ and $T(v_2) = (-1, 1)$. Determine a formula for $T(x_1, x_2)$, and use that formula to find $T(2, -3)$.